MIL-P-80135B
17 November 1987
SUPERSEDING
MIL-P-80135A
16 November 1973

MILITARY SPECIFICATION

PRESSES, MECHANICAL STRAIGHT SIDED, TIE RODS, MULTIPLE POINT, NON-GEARED TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers non-geared, flywheel, vertical, straight sided, tie rod type presses with single action, and multiple point connection with equipment and accessories specified herein.

1.2 Classification. The presses shall be of the following sizes. The size to be furnished shall be as specified (see 6.2.1).

Size 1 - 20 ton capacity.
Size 2 - 30 ton capacity.
Size 3 - 45 ton capacity.
Size 4 - 50 ton capacity.
Size 5 - 60 ton capacity.
Size 6 - 75 ton capacity.
Size 7 - 100 ton capacity.
Size 8 - 125 ton capacity.
Size 9 - 150 ton capacity.
Size 10 - 200 ton capacity.
Size 11 - 250 ton capacity.
Size 12 - 300 ton capacity.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, Memphis, TN 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATION

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Service.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

2.1.2 Other Government document. The following other Government document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be that in effect on the date of the solicitation.

U. S. DEPARTMENT OF LABOR

CFR 29, CHAPTER XVII, PART 1910 - Occupational Safety and Health Standards.

(Copies of specifications, standards, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment document which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B5.1 - T-Slots, Their Bolts, Nuts, and Tongues.

ANSI B11.1 - Mechanical Power Presses, Safety Requirements for the Construction, Care, and Use.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018-3363).

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103-1187).

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery.

(Nongovernment standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Design. The press shall be new and one of the manufacturer's current models capable of pressing operations in accordance with the requirements specified herein. The press shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2.1 Dimensional system. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the press. When only one dimensional system is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all dimensions, measurements, sizes, and capacities are given in US units. These dimensions may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.1.1 Measuring and indicating device calibrations. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices such as depth stops, pressure gauges, and other similar devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the press shall be graduated in the same system.

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3.2.2 Reclaimed materials. The press may contain reclaimed materials provided such materials will not jeopardize the intended use and performance of the press. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the press.

3.2.3 Energy efficiency. The press and its components that directly consume energy in normal operation shall be designed and constructed for the highest degree of energy efficiency as governed by the latest developments available within the industry.

3.2.4 Controls. All operating controls shall be located conveniently to the operator at his normal work station.

3.2.5 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the press that present safety hazards. The safety devices shall not interfere with the operation of the press. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. All parts, components, mechanisms, and assemblies furnished on the press, whether or not specifically required herein, shall comply with all of the requirements of CFR 29, Chapter XVII, Part 1910 and ANSI B11.1 that are applicable to the press itself. The press shall be equipped with all point-of-operation guarding normally furnished as standard on the manufacturer's commercial press supplied to the commercial market. If specific point-of- operation guarding is required, the procuring activity should specify the exact configuration of the guard required because the guard configuration is dependent on the size and configuration of the workpieces. The above, and any other additional safety and health requirements should be specified in detail (see 6.2.1).

3.2.6 Mercury restriction. The press shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the press.

3.2.8 Environmental protection. The press shall be so designed and constructed that, under the operating, service, transportation, and storage conditions described herein, the press shall not emit materials hazardous to the ecological system as prohibited by Federal, state, or local statutes in effect at the point of installation.

3.2.9 Lubrication. An automatic recirculating lubricating system shall be provided for all parts and mechanisms requiring a continuous flow of lubricating oil. Means for determining oil level in each reservoir shall be by an oil level sight gauge, a low oil level warning light, or a similar device. Mechanisms requiring periodic manual lubrication shall be readily accessible for servicing. Each recirculating system shall include a cleanable or replaceable filter.

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3.2.10 Interchangeability. All parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or press.

3.3 Construction. The press shall be constructed of parts which are new, without defects, and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the press at its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects, scale, and mismatching. Processes such as welding, peening, plugging, or filling with solders or pastes shall not be used on castings or forgings to reclaim any defective components for use on the press. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.4 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the press shall be painted in accordance with the manufacturer's standard commercial practice and color.

3.3.6 Threads. All threaded parts used on the press and its related attachments and accessories shall conform to FED-STD-H28.

3.3.7 Electromagnetic interference control. When specified (see 6.2.1), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystems class and the emission and susceptibility requirement shall be as specified.

3.4 Components. The press shall consist of not less than a bed, two uprights, a crown, and all components essential for the operation of the press.

3.4.1 Frame. The frame shall consist of a bed, uprights, and crown. The frame shall be manufactured of material that will provide sufficient strength and rigidity to permit operation of the press at its maximum rated tonnage capacity without damage or permanent deformation. All mating surfaces of the press shall be machined and stress relieved to assure proper alignment of the bed, uprights, and crown.

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3.4.1.1 Bed. The bed design, material, and construction shall provide the strength and rigidity to resist deflection when operating the press at its maximum rated tonnage. Not less than four hold down points shall be provided for anchoring the press to a foundation.

3.4.1.2 Uprights. The uprights shall be of sufficient cross sectional area to resist deflection of the press during operation. The press shall maintain squareness to the bed and crown, right to left, and front to back, by means of keys installed between the uprights and bed and the uprights and crown.

3.4.1.3 Crown. The crown shall be of box-type construction and braced to provide the necessary strength and rigidity to minimize deflection. The crown shall support all parts and mechanisms associated with the press drive.

3.4.1.4 Tie rods. Four tie rods shall secure the press bed, uprights, and the crown in position. Elongation of the tie rods shall be of the manufacturer's standard design. When a specific means for tie rods elongation is required, the procurring activity shall specify and fully describe, the elongation method (see 6.2.1). The tie rods shall pre-load the press to twice the rated capacity.

3.4.2 Non-geared drive. The drive shall be the non-geared type. Provisions shall be provided for reversing the direction of the drive. When specified (see 6.2.1), a variable speed drive shall be provided.

3.4.2.1 V-belts. The V-belts shall be of the type and size to transmit the required power without slippage. Belts, shall be provided in matched sets for maximum power transmission. Means shall be provided for adjusting and replacing the belts.

3.4.2.2 Flywheel. The flywheel shall be balanced after machining, and designed with mass and speed to supply kinetic energy required to perform any operation within the capacity of the press without imposing an overload condition on the drive motor or other driving components. When specified (see 6.2.1), a flywheel brake shall be provided for stopping the flywheel within 30 seconds after the drive motor power has been turned off.

3.4.2.3. Clutch. The clutch shall be of the air engaged and spring released type. The engagement and disengagement of the clutch shall be synchronized with the brake so that when the clutch is engaged, the brake is released, and when the clutch is disengaged, the brake shall be applied. The clutch control system shall automatically release the clutch and apply the brake in the event of failure in either the air or electrical supply or both.

3.4.2.4 Brake. The brake shall be of the type that is held disengaged by air pressure, and is automatically applied by spring pressure when air pressure is released. The brake application shall be synchronized with operation of the clutch (see 3.4.2.3). When the control is set for single stroke operation, the brake shall stop the shaft when the slide is at or near the top of the stroke.

3.4.2.5 Main shaft. The shaft shall be of the size, material, and strength to transmit maximum imposed torsional loads without permanent deformation or

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deflection. The shaft shall support combinations of flywheel, clutch, brake, V-belt drive surfaces, and auxiliary units such as feed control. The shaft shall be a configuration with two or more throws or eccentrics to convert rotary to reciprocating motion of the slide. The shaft shall be supported by and revolve in replaceable bearings. Shaft bearings shall be line-reamed and finish-fitted to provide a running clearance in accordance with the minimum requirements of the size shaft furnished. When specified (see 6.2.1), a shaft extension shall be provided as a power source for driving auxiliary equipment.

3.4.2.6 Connecting rod. The connecting rod shall be of the mass required to transmit the maximum rated tonnage without permanent deformation or deflection. The shaft end of the connecting rod shall be of the split cap design, and shall be fitted to the shaft with replaceable bronze bushings. The slide end of the connecting rod shall be self-aligning, and shall be fitted to the slide in accordance with the manufacturer's standard commercial practice.

3.4.3 Slide. The slide shall be constructed with ways and gibs located at each corner. The design and construction of the slide, ways, and gibs shall provide precise alignment of the slide with the bed at all times, under all operating conditions. At least two corners of the slide shall have adjustable gibs for precision adjustment of alignment and to compensate for wear. The ways and gibs shall be of sufficient length to maintain precise control of the slide throughout its maximum stroke length. When specified (see 6.2.1), the die area of the slide face shall be T-slotted in accordance with ANSI B5.1. Provisions shall be made for installation of crossbar knockouts.

3.4.3.1 Slide adjustment. The slide shall be adjustable vertically to change its position with relation to the bolster plate or bed, when the slide is at the bottom of its stroke. The amount of adjustment shall be not less than that specified in table I. Unless otherwise specified (see 6.2.1), presses with a rated tonnage capacity of 150 tons or less may be adjusted manually. Presses rated at more than 150 tons shall have an electric or air motor driven adjusting mechanism. Each press shall be equipped with a slide position indicator to show the position of the slide at all points of the stroke.

3.4.4 Counterbalance. Each press shall be equipped with an adjustable pneumatic counterbalance system for offsetting the weight of the slide and of the dies. The system shall include valving, gages, a pressure regulator to control air pressure for compensating varying die weights, and means for lubricating the cylinders.

3.4.5 Bolster. Unless otherwise specified (see 6.2.1), the manufacturer's standard bolster for the press specified shall be furnished. When T-slotted bolsters are furnished, slotting shall be in accordance with ANSI B5.1.

3.4.6 Die cushion. When specified (see 6.2.1), a die cushion shall be furnished, in accordance with the manufacturers standard design.

3.4.7 Peak load measurement system. When specified (see 6.2.1), the press shall be furnished with a load measurement system that measures and provides a visual indication of the peak load exerted by the press slide during a pressing cycle.

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TABLE I. Sizes and capacities.

Size	1	2	3	4	5	6	7	8	9	10	11	12
Capacity, nominal tons	20	30	45	50	60	75	100	125	150	200	250	300
Effective die dimensions (inches) 1/	24x18 30x18 36x18	20x18 24x18 30x18 36x18	24x24 30x24 36x24 42x24 48x24	21x24 30x24 30x30 36x18 36x24 36x30 42x18 42x24 42x30 48x18 48x24 48x30 60x24 60x30 72x24	24x24 30x24 30x18 36x24 36x30 42x18 42x24 42x30 48x18	21x24 30x30 30x48 30x54 36x18 36x24 36x30 42x18 42x24 42x30 42x36 48x24 48x30 48x36 60x30 60x36 60x42 60x48 72x30 72x36 84x30	24x30 30x60 36x24 36x30 36x36 42x24 42x30 42x36 48x24 48x30 48x36 48x42 48x48 60x30 60x36 60x42 72x36 72x48 84x30	36x24 36x30 36x36 42x24 42x30 42x36 48x24 48x30 48x36 60x36 72x36 84x36 96x36 120x36	36x24 30x30 36x30 36x36 42x24 42x30 42x36 48x24 48x30 48x36 48x42 48x48 60x36 60x42 60x48 72x36 72X42 72x48 84x36 96x36 120x36	42x36 42x42 48x42 60x42 60x48 72x42 72x48 72x42 84x36 96x42 120x42	48x42 60x42 60x48 72x36 72x42 72x48 84x42	48x42 60x42 60x48 72x42 72x48 84x48
Stroke length, standard	1	2	2	3	3	4	4	4	4	4	8	8
Strokes per minute, standard	100	100	100	90	90	70	70	70	65	65	60	60
Slide adjustment	2	2	3	3	3	3	4	4	6	6	8	8
Shutheight 2/	11	11	13	13	13	15	16	19	20	22	22	22

Requirement shall be not less than those shown.

1/ Dimensions of bed, bolster, or slide face used to determine effective die area.

2/ Vertical space available for die, from the slide to bed with stroke down and slide adjustment up (S.D.A.U).

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3.4.8 Electrical system. Unless otherwise specified (see 6.2.1), the electrical system shall conform to NFPA 79. Each press shall be wired to draw all of its electrical power from a single 230/460-volt, 3-phase, 60-Hz circuit. Each press shall have a fused safety disconnect switch or circuit breaker. The press shall be initially wired for operation on 460 volts. An identified terminal for grounding the press when it is installed shall be mounted in or near the disconnect switch. The terminal shall be suitable for connecting the size grounding conductor specified in NFPA 79 for the disconnect fuse rating.

3.4.8.1 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2.1), each motor shall have a dripproof enclosure.

3.4.8.2 Solid state components. Solid state design shall be used throughout for electronic components.

3.4.8.3 Control circuit voltage. The circuits shall be isolated from the input electric power supply by a low-voltage transformer having a secondary voltage no greater than 120 volts.

3.5 Size and capacity. Unless otherwise specified (see 6.2.1), the size and capacity of the press shall be not less than those specified in table I for the press size specified.

3.6 Performance. Each press shall be capable of performing metal working operations within the rated tonnage as specified herein. Maximum tonnage rating shall be developed when the slide face is 1/16 inch or more from the bottom of the stroke.

3.6.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2.1).

3.7 Alignment accuracy. The bottom surface of the slide shall be parallel to the top surface of the bed within the following accuracies:

a. 0.001 inch per foot at bottom of stroke.
b. 0.003 inch per foot at midstroke.

3.8 Standard equipment. Unless otherwise specified, all equipment normally furnished with the manufacturer's standard commercial press shall be furnished. When specified (see 6.2.1), required additional equipment shall be furnished and fully described.

3.9 Optional equipment. Optional equipment shall be furnished as specified and shall be fully described (see 6.2.1).

3.10 Repair parts. Repair parts shall be furnished as specified (see 6.2.1).

3.11 Marking on plates and charts. All words on plates and charts shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background.

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3.11.1 Lubrication chart or plate. A lubrication chart or plate shall be securely attached to each press. If a chart is furnished, it shall be placed in a transparent plastic folder or permanently sealed between clear plastic sheets with suitable means of mounting. The chart or plate shall contain the following information:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity

3.11.2 Nameplate. A nameplate shall be securely attached to each press. The nameplate shall contain the information listed below. If the press is a special model, the model designation shall include the model of the basic standard press and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Air input requirements
Capacity, tons maximum, rated
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.12 Hour meter. Unless otherwise specified (see 6.2.1), the press shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 99,999 hours in increments of not greater than one hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the press. Upon reaching 99,999 hours, the meter readout shall automatically return to zero and continue to accumulate time.

3.13 Technical data. When technical data is required it shall be furnished in accordance with the requirements of the contract. All technical data furnished shall be written in the English language.

3.14 Workmanship. Workmanship of the press and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspections. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government re-

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serves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections, tests, and examinations shall be performed in the manufacturer's designated indoor test area under the ambient temperature, the relative humidity, and the air pressure existing inside the building at the time the inspections, tests, and examinations are performed.

4.4 First article inspection. When first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. Failure of the item to pass the first article examination and all tests shall be cause for rejection.

4.5 Quality conformance inspection. A quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), the quality conformance inspection shall consist of the examination in 4.6, the test in 4.7.1, and the inspection in 4.8. Failure of the item to pass the examination, the test, and the inspection shall be cause for rejection.

4.6 Examination. The press shall be visually examined to determine compliance with all requirements of this specification.

4.7 Tests.

4.7.1 Operational test. The press shall be operated at no load for not less than 30 minutes. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.7.2 Alignment tolerance test. Each press shall be tested to determine that it complies with the alignment tolerance requirements of 3.7.

4.7.3 Performance tests. Each press shall be tested by applying a load equal to the press tonnage rating, plus 10 percent, between the slide face and

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the bolster. The load may be applied using hydraulic jacks equipped with calibrated pressure gauges, or any other similar device that accurately applies and measures the applied load. After completion of the load test, the press shall be visually examined, without dismantling, for damage to components, adjustments, and alignment.

4.7.4 Electromagnetic interference control tests. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.7.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-M-18058.

6. NOTES

6.1 Intended use. Presses specified herein are intended for blanking, trimming, piercing, drawing, and forming operations with metal. The presses are intended for prototype, short-run or production operations.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Press size required (see 1.2).

c. First article, if required (see 3.1).

d. If press design and construction is required in a specific dimensional system (US or SI), state required system (see 3.2.1).

e. If measuring and indicating devices are required to be graduated in a specific measurement system (US or SI), state required system (see 3.2.1.1).

f. Additional safety and health requirements, if required (see 3.2.5).

g. Painting, if different (see 3.3.5).

h. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.7).

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i. Tie rod elongation, if different, fully describe (see 3.4.1.4).

j. Variable speed drive, if required (see 3.4.2).

k. Flywheel brake, if required (see 3.4.2.2).

l. Drive shaft extension, if required (3.4.2.5).

m. T-slots in slide, if required (see 3.4.3).

n. Power slide adjustment, for presses rated at 150 ton or less, if required (see 3.4.3.1).

o. Bolster, if different (see 3.4.5).

p. Die cushion, if required (see 3.4.6).

q. Peak load measurement system, if required (see 3.4.7).

r. Electrical system and input power, if different (see 3.4.8).

s. Motor enclosure, if different (see 3.4.8.1).

t. Size and capacity, if different (see 3.5).

u. Additional performance features, if required (see 3.6.1).

v. Standard equipment, if different (see 3.8).

w. Specify and fully describe optional equipment, if required (see 3.9).

x. Specify applicable repair parts, if required (see 3.10).

y. Hour meter, if different (see 3.12).

z. First article inspection, if different (see 4.4).

aa. Quality conformance inspection, if different (see 4.5).

bb. If preservation, packing, and marking in accordance with MIL-M-18058 is required, specify level of preservation packing required (see 5.1).

6.2.2 Contract data requirements. Required technical data such as operator's manuals, parts lists, wiring diagrams, foundation and anchor bolt plans, and acceptance test reports should be specified on a DD Form 1423, Contract Data Requirements List, incorporated into the contract.

6.3 First article When first article inspection is required, the item to be tested should be the first offered for acceptance under the contract.

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The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Cross-reference of classification changes. The press classification changes from the previous issue of this specification are as follows:

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20	Size 1 - 20 ton capacity
30	Size 2 - 30 ton capacity
45	Size 3 - 45 ton capacity
50	Size 4 - 50 ton capacity
60	Size 5 - 60 ton capacity
75	Size 6 - 75 ton capacity
100	Size 7 - 100 ton capacity
125	Size 8 - 125 ton capacity

MIL-P-80135A	MIL-P-80135B
150	Size 9 - 150 ton capacity
200	Size 10 - 200 ton capacity
250	Size 11 - 250 ton capacity
300	Size 12 - 300 ton capacity

6.5 Subject term (key word) listing.

Mechanical Press
Metal working
Pressing
Punch press

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - AL
Navy - SH
Air Force - 99

Preparing activity:
DLA - IP

(Project 3443-0085)

Review activity:
Air Force - 84

User activities:
Army - AV
Navy - MC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-80135B	Presses, Mechanical Straight Sided, Tie Rods, Multiple Point, Non-Geared Type

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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